

Problem Set

1. Prove that the Diophantine equation $x^2 + y^2 = 2023$ has no integer solutions.
2. Let $n = a^2 + b^2$ for some integers a, b such that $\gcd(a, b) = 1$. Prove that every positive divisor d of n can itself be written as the sum of two coprime integer squares. (*Hint: Utilize the $4k + 3$ prime constraint*).
3. Prove that the Diophantine equation $y^2 = x^3 + 7$ has no integer solutions. (*Hint: Rewrite the equation as $y^2 + 1 = x^3 + 8$, factor the right side, and analyze the prime factors of the quadratic factor using the $4k + 3$ lemma*).
4. Let p be a prime such that $p \equiv 3 \pmod{4}$. Prove that the negative Pell equation $x^2 - py^2 = -1$ has no integer solutions.
5. Let n be a square-free positive integer. Prove that the circle given by the equation $x^2 + y^2 = n$ contains at least one point with rational coordinates $(x, y) \in \mathbb{Q}^2$ if and only if no prime divisor of n is congruent to $3 \pmod{4}$.
6. Let $r_2(n)$ denote the number of pairs of integers (x, y) such that $x^2 + y^2 = n$. Let $n = p_1^{a_1} \dots p_k^{a_k} q_1^{b_1} \dots q_m^{b_m}$ be the prime factorization of n , where $p_i \equiv 1 \pmod{4}$ and $q_j \equiv 3 \pmod{4}$. Prove that if any exponent b_j is odd, then $r_2(n) = 0$.
7. Let $P(n)$ and $Q(n)$ be polynomials with integer coefficients. Prove that there does not exist any integer n that satisfies the equation:

$$P(n)^2 + Q(n)^2 = 4n + 3.$$
 (*Hint: Consider the prime factorization of integers of the form $4k + 3$*).
8. Let $k \geq 2$ be an integer. Prove that the Diophantine equation $x^2 + y^2 = z^k$ has no solutions in positive integers (x, y, z) satisfying both $\gcd(x, y) = 1$ and $z \equiv 3 \pmod{4}$.
9. Let p be a prime such that $p \equiv 3 \pmod{4}$. If n is a positive integer such that $p \mid n$ but $p^2 \nmid n$, prove that n cannot be expressed as the sum of two integer squares.

- 10.** A classic proof that there are infinitely many primes of the form $4k + 1$ involves showing that for any integer N , the prime factors of $P(N) = (N!)^2 + 1$ must be of the form $4k + 1$. Explain rigorously why constructing an analogous polynomial $Q(N) = (N!)^2 + 3$ completely fails to prove that there are infinitely many primes of the form $4k + 3$.